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Module 1: ES6 Fundamentals

Objective: Develop a deep understanding of ES6 features essential for modern JavaScript and React development.

Topics Covered:

- Function vs block scope
- Arrow functions
- Template literals
- Object destructuring
- Spread vs rest operators
- Array methods: map, filter, reduce
- Classes and inheritance
- Promises, async/await

- try, catch, finally
- Method binding

Module 2: Pre-requisites & Environment Setup

Objective: Build a strong technical foundation by setting up a modern development environment and mastering essential TypeScript fundamentals.

Topics Covered:

- Installing Node.js (LTS), npm, pnpm/yarn
- VS Code setup with essential extensions
- TypeScript basics: types, inference, interfaces vs types
- Type safety in functions & components
- Enterprise-grade folder structure for Next.js apps

Module 3: React Fundamentals

Objective: Develop a deep understanding of React's core building blocks to create scalable, component-driven interfaces.

Topics Covered:

- React's role in modern web development
- Functional components & JSX syntax
- Props, State, Event Handling
- Rendering lists & keys
- Conditional rendering patterns
- Controlled vs uncontrolled components
- Hooks: useState, useEffect
- Component composition & reusability

Module 4: Next.js Fundamentals – App Router

Objective: Gain mastery over Next.js architecture and routing to build fast, scalable full-stack applications.

Topics Covered:

- Next.js vs React
- App Router architecture

- File-based routing (nested routes, layouts)
- Client Components vs Server Components
- Static vs Dynamic Rendering

Module 5: Server Actions in Next.js

Objective: Learn how Server Actions streamline backend logic and replace API routes for secure data handling.

Topics Covered:

- Introduction to Server Actions
- Replacing/augmenting API routes
- Creating & using Server Actions
- Form submissions with validations
- Error handling
- **Practical:** Build “Contact Us” form storing data via Server Actions + API Route

Module 6: State Management & Data Fetching

Objective: Master client/server data flow and implement robust state management patterns for enterprise apps.

Topics Covered:

- When to use client vs server components
- useState, useReducer, Context API
- Server-side & client-side data fetching
- SSR (Server-Side Rendering)
- Optimistic UI updates
- **Practical:** Product List + Add-to-Cart with server integration

Module 7: Enterprise-Grade Architecture Patterns

Objective: Apply real enterprise patterns to structure scalable, maintainable Next.js applications.

Topics Covered:

- MVVM / Clean Architecture design
- Module-based folder structure

- TypeScript best practices
- Form management: React Hook Form / Formik
- Schema validation: Zod / Yup
- Error boundaries & global loading states
- Internationalization (i18n)
- **Practical:** Multi-language dashboard module

Module 8: Performance Optimization & Production Readiness

Objective: Optimize applications for speed, security, and reliability in real-world production environments.

Topics Covered:

- Code splitting & dynamic imports
- Server components: advantages & limitations
- Performance measurement with Lighthouse
- Security best practices (XSS, CSRF, SSRF)
- Environment variables & configuration strategies

Module 9: Capstone Project – Enterprise Order Management Portal

Objective: Consolidate all concepts into a real-world enterprise solution built end-to-end.

Project Features:

- Product catalog with search & filtering
- Server Action-based Order Creation Form
- Order listing with pagination & sorting
- Multi-language support

Tech Stack:

Next.js (App Router), React, TypeScript, Tailwind CSS, MySQL

Module 10: Understanding Customer's Node.js Dependencies

Objective: Examine and understand the Node.js modules included in the customer's shared package.json for effective integration and usage.

Topics Covered:

- AG Grid — data-intensive grid systems

- Chart.js — visual chart rendering
- Zustand — lightweight global state management
- xlsx — Excel file parsing and generation

